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CANDIDATE NUMBER

SYDNEY GRAMMAR SCHOOL



2016 Half-Yearly Examination

FORM VI

MATHEMATICS EXTENSION 1

Tuesday 1st March 2016

General Instructions

- Writing time — 1 hour and 30 minutes
- Write using black pen.
- Board-approved calculators and templates may be used.

Total — 55 Marks

- All questions may be attempted.

Section I – 7 Marks

- Questions 1–7 are of equal value.
- Record your answers to the multiple choice on the sheet provided.

Section II – 48 Marks

- Questions 8–11 are of equal value.
- All necessary working should be shown.
- Start each question in a new booklet.

Collection

- Write your candidate number on each answer booklet and on your multiple choice answer sheet.
- Hand in the booklets in a single well-ordered pile.
- Hand in a booklet for each question in Section II, even if it has not been attempted.
- If you use a second booklet for a question, place it inside the first.
- Write your candidate number on this question paper and hand it in with your answers.
- Place everything inside the answer booklet for Question Eight.

Checklist

- SGS booklets — 4 per boy
- Multiple choice answer sheet
- Reference sheet
- Candidature — 111 boys

Examiner

DWH

SECTION I - Multiple Choice

Answers for this section should be recorded on the separate answer sheet handed out with this examination paper.

QUESTION ONE

1

The value of $\sin^{-1}\left(\frac{\sqrt{3}}{2}\right)$ is:

(A) $\frac{\pi}{2}$

(B) $\frac{\pi}{3}$

(C) $\frac{\pi}{4}$

(D) $\frac{\pi}{6}$

QUESTION TWO

1

Find $\int \frac{1}{\sqrt{25-x^2}} dx$.

(A) $\sin^{-1}\left(\frac{x}{5}\right) + C$

(B) $\frac{1}{5} \sin^{-1}\left(\frac{x}{5}\right) + C$

(C) $\sin^{-1}(5x) + C$

(D) $\frac{1}{5} \sin^{-1}(5x) + C$

QUESTION THREE

1

Evaluate $\int_0^{\frac{\pi}{3}} \sec^2 x \, dx$.

(A) $\frac{1}{\sqrt{3}}$

(B) $\frac{2}{\sqrt{3}}$

(C) $\sqrt{3}$

(D) $2\sqrt{3}$

QUESTION FOUR

1

Which of the following is a point on the parabola $x^2 = 4ay$?

(A) $S(0, a)$

(B) $S'(0, -a)$

(C) $R\left(\frac{2a}{r}, \frac{a}{r^2}\right)$

(D) $Q(aq^2, 2aq)$

QUESTION FIVE

1

The area of a sector in a circle is given by the formula $A = \frac{1}{2}r^2\theta$. The radius r is fixed but the angle θ is increasing at a constant rate. The rate of change of A is:

(A) constant.

(B) zero.

(C) decreasing.

(D) increasing.

QUESTION SIX

1

The identity $\sin^{-1}(\sin x) = x$ is:

- (A) false for all real x .
- (B) true for all real x .
- (C) false for $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$.
- (D) true for $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$.

QUESTION SEVEN

1

Which of the following expressions is identical to $\sin\left(\frac{3\pi}{2} - x\right)$?

- (A) $\cos\left(\frac{3\pi}{2} + x\right)$
- (B) $\sin\left(\frac{3\pi}{2} + x\right)$
- (C) $-\cos\left(\frac{3\pi}{2} + x\right)$
- (D) $-\sin\left(\frac{3\pi}{2} + x\right)$

————— End of Section I —————

Exam continues next page ...

SECTION II - Written Response

Answers for this section should be recorded in the booklets provided.

Show all necessary working.

Start a new booklet for each question.

QUESTION EIGHT (12 marks) Use a separate writing booklet.	Marks
(a) Evaluate $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} 3 \sin 2x \, dx$.	2
(b) Find $f'(x)$ given $f(x) = \sin^2 x$.	1
(c) Solve the equation $2 \cos^2 \theta + \sin \theta = 1$ for $0 \leq \theta \leq 2\pi$.	3
(d) Consider the function $y = 3 \sin^{-1} \left(\frac{x}{2} \right)$.	
(i) Sketch the graph of the function.	2
(ii) Find the gradient of the curve at the point where $x = 1$.	2
(e) Find the acute angle between the lines with equations $y = -\frac{5}{2}x + 2$ and $y = -\frac{2}{3}x + 5$. Express your answer correct to the nearest minute.	2

QUESTION NINE (12 marks) Use a separate writing booklet.

Marks

- (a) A spherical balloon is being inflated with air such that the volume V of the balloon increases at a constant rate of 10 cm^3 per second.

The volume of a sphere of radius r is given by $V = \frac{4}{3}\pi r^3$.

- (i) Find an expression for the rate of change of the radius r of the balloon. 2

- (ii) What is the radius at the instant when the radius is increasing at 0.2 cm per second? Give your answer correct to the nearest millimetre. 1

- (b) Find the general solution to the equation $\sin x = \frac{1}{\sqrt{2}}$. Give your solution in radians in exact form. 2

- (c) Consider the equation $3 \cos x + 4 \sin x = 4$.

- (i) Write $3 \cos x + 4 \sin x$ in the form $R \cos(x - \alpha)$, where $0 < \alpha < 90^\circ$ and $R > 0$. Give α correct to the nearest degree. 2

- (ii) Hence solve $3 \cos x + 4 \sin x = 4$ in the domain $0^\circ \leq x \leq 360^\circ$. Give your solutions correct to the nearest degree. 2

- (d) Let $t = \tan\left(\frac{7\pi}{8}\right)$.

- (i) Using a t -substitution for the expression $\tan\left(\frac{7\pi}{4}\right)$, or otherwise, show that 1

$$t^2 - 2t - 1 = 0.$$

- (ii) Hence find the exact value of $\tan\left(\frac{7\pi}{8}\right)$. 2

QUESTION TEN (12 marks) Use a separate writing booklet.

Marks

- (a) (i) Use the properties of logarithms to write $\log_e \left(\frac{(x+1)^2}{2x} \right)$ without fractions. 1

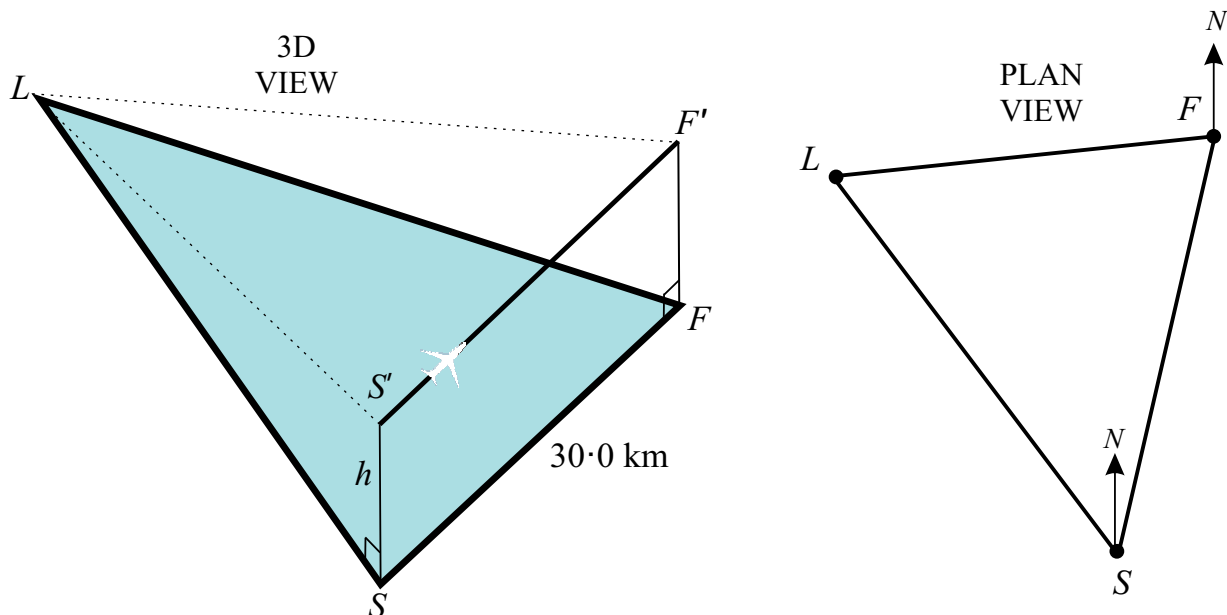
- (ii) Find the coordinates of the stationary point on the curve $y = \log_e \left(\frac{(x+1)^2}{2x} \right)$. 2

- (b) Use Mathematical Induction to prove that for all positive integers n , 3

$$(1 \times 4) + (2 \times 5) + (3 \times 6) + \cdots + (n \times (n+3)) = \frac{1}{3}n(n+1)(n+5).$$

- (c) The parabola $x^2 = 6y$ intersects with the line $y = 2 - x$ at two points, A and B . The tangents to the parabola at A and B intersect at P . Find the co-ordinates of P . 2
(You may use the chord of contact formula which is on the Reference Sheet.)

- (d)



A pilot flew a plane over a flat region at a constant altitude h . When she was at point S' vertically above point S , she observed a landmark L on the ground on a bearing of $323^\circ T$ at an angle of depression of 7° . After flying 30.0 kilometres on a bearing of $013^\circ T$ she arrived at point F' vertically above F . She observed the same landmark L on a bearing of $264^\circ T$.

- (i) Show that $\angle FLS = 59^\circ$. 1
 (ii) Find LS in terms of h . 1
 (iii) Find her altitude h . Give your answer correct to the nearest 100 metres. 2

QUESTION ELEVEN (12 marks) Use a separate writing booklet.

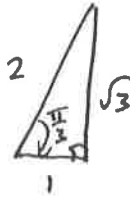
Marks

- (a) Suppose $P(2ap, ap^2)$ is a point on the parabola $x^2 = 4ay$.
- (i) Show that the equation of the normal to the parabola at the point P is **2**
 $x + py = 2ap + ap^3$.
 - (ii) The normal at P cuts the y -axis at Q . The point R moves such that Q is the midpoint of RP . Show that the co-ordinates of R are $(-2ap, 4a + ap^2)$. **2**
 - (iii) Find the Cartesian equation of the locus of R as P varies. **2**
 - (iv) Describe the locus of R as a transformation of the original parabola $x^2 = 4ay$. **1**
- (b) The curve $y = -\log_e(x + 1)$, the y -axis and the line $y = 1$ enclose a region.
- (i) Sketch this region, clearly showing any intercepts with the axes and asymptotes. **2**
 - (ii) This region is rotated around the y -axis to form a solid of revolution. Find the volume of this solid. **3**

————— End of Section II —————

END OF EXAMINATION

(1)



$$\sin^{-1}\left(\frac{\sqrt{3}}{2}\right) = \frac{\pi}{3} \quad (B)$$

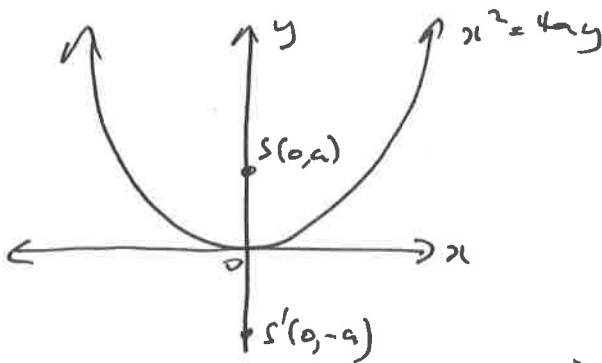
(2)

$$\int \frac{1}{\sqrt{25-x^2}} dx = \sin^{-1}\left(\frac{x}{5}\right) + C \quad (A)$$

(3)

$$\int_0^{\pi/3} \sec^2 x dx = [\tan x]_0^{\pi/3} = \tan \frac{\pi}{3} - 0 = \sqrt{3} \quad (C)$$

(4)



$$R: x^2 = \left(\frac{2a}{r}\right)^2 = \frac{4a^2}{r^2} = 4a\left(\frac{a}{r^2}\right) = 4ay \quad \checkmark$$

$$Q: x^2 = (a_1^2)^2 \neq 4a(2a_2) \quad \times$$

$\therefore (C)$

(5)

r is constant

$$A = \frac{1}{2} r^2 \theta$$

$$\frac{dA}{d\theta} = \frac{1}{2} r^2 \frac{d\theta}{d\theta}, \text{ which is constant } (A)$$

(6)

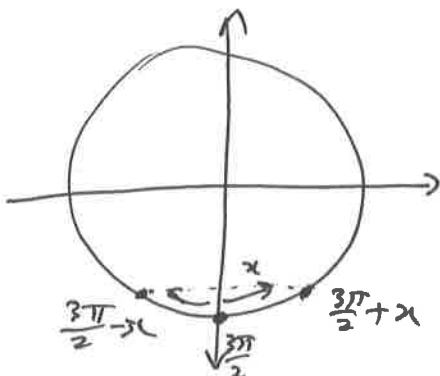
$$\sin^{-1}(\sin x) = x$$

~~range~~ of $\sin^{-1}(y)$ is $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$
domain

$\therefore$ range of RHS is also $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$

$\therefore (D)$

(7)



$\frac{3\pi}{2} - x$ and $\frac{3\pi}{2} + x$ have equal
sine values

$\therefore (B)$

8 (a) $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} 3 \sin 2x \, dx = 3 \left[-\frac{\cos 2x}{2} \right]_{\frac{\pi}{6}}^{\frac{\pi}{3}}$ ✓

$$= -\frac{3}{2} \left[\cos \frac{2\pi}{3} - \cos \frac{\pi}{3} \right]$$

$$= -\frac{3}{2} \left[-\frac{1}{2} - \frac{1}{2} \right]$$

$$= \frac{3}{2}.$$
 ✓

(b) $f(x) = \sin^2 x$

$f'(x) = 2 \sin x \cos x$ (Chain rule) ✓

(c) $2 \cos^2 \theta + \sin \theta = 1$ $0 \leq \theta \leq 2\pi$

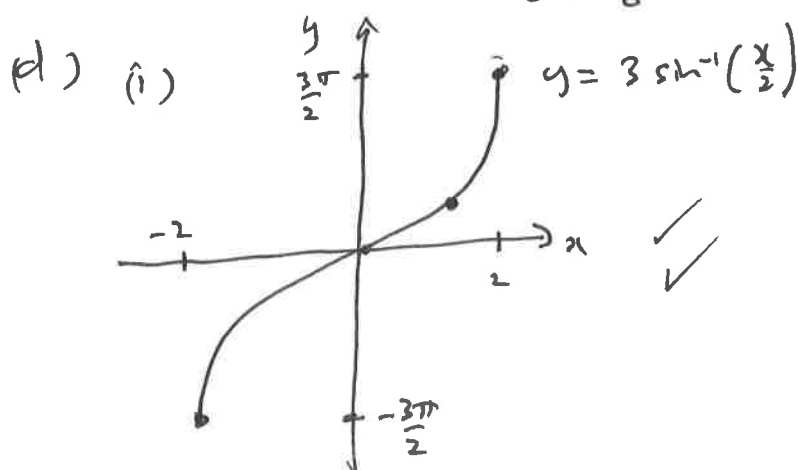
$2(1 - \sin^2 \theta) + \sin \theta = 1$ ✓

$0 = 2 \sin^2 \theta - \sin \theta - 1$

$= (2 \sin \theta + 1)(\sin \theta - 1)$ ✓

$\therefore \sin \theta = -\frac{1}{2} \text{ or } 1$ ✓

$\theta = \frac{\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}$ ✓



$-1 \leq \frac{x}{2} \leq 1$

$-2 \leq x \leq 2$

$-\frac{\pi}{2} \leq \sin^{-1}\left(\frac{x}{2}\right) \leq \frac{\pi}{2}$

$-\frac{3\pi}{2} \leq 3 \sin^{-1}\left(\frac{x}{2}\right) \leq \frac{3\pi}{2}$

(ii) $y = 3 \sin^{-1}\left(\frac{x}{2}\right)$

$\frac{dy}{dx} = 3 \cdot \frac{1}{\sqrt{1 - \left(\frac{x}{2}\right)^2}} \cdot \frac{1}{2}$ ✓

when $x = 1$

$m_{\text{tangent}} = \frac{3}{2\sqrt{\frac{3}{4}}} = \sqrt{3}$ ✓

8(e) $y = -\frac{5}{2}x + 2$ $m_1 = -\frac{5}{2}$ Let $\theta = \angle$ between the two lines
 $y = -\frac{2}{3}x + 5$ $m_2 = -\frac{2}{3}$

$$\tan \theta = \left| \frac{-\frac{5}{2} - (-\frac{2}{3})}{1 + (-\frac{5}{2})(-\frac{2}{3})} \right|$$

$$= \frac{11}{16}$$

$$\theta \approx 34^\circ 31'$$

$\therefore$ The angle between the two lines is $\approx 34^\circ 31'$

(9) (a) (i) $\frac{dV}{dt} = +10 \text{ cm}^3/\text{s}$

$$V = \frac{4}{3}\pi r^3 \text{ cm}^3$$

$$\frac{dV}{dr} = 4\pi r^2 \text{ cm}^2/\text{cm}$$

$$\frac{dr}{dt} = \frac{dr}{dV} \cdot \frac{dV}{dt}$$

$$= \frac{1}{4\pi r^2} \cdot 10 \text{ cm/s}$$

$$= \frac{5}{2\pi r^2} \text{ cm/s}$$

(ii) when $\frac{dr}{dt} = 0.2 \text{ cm/s}$

$$0.2 = \frac{5}{2\pi r^2}$$

$$r^2 = \frac{25}{2\pi}$$

$$r = \sqrt{\frac{25}{2\pi}} \approx 2.0 \text{ cm}$$

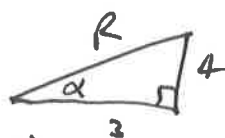
(9)(b) $\sin x = \frac{1}{\sqrt{2}}$ ✓ ✓

$x = \frac{\pi}{4} (-1)^n + \pi n$ where n is an integer

or $x = \frac{\pi}{4} + 2\pi n$ or $\frac{3\pi}{4} + 2\pi n$ where n is an integer

(c)(i) $3\cos x + 4\sin x = R \left(\frac{3}{R} \cos x + \frac{4}{R} \sin x \right)$

so if $\frac{3}{R} = \cos \alpha$ and $\frac{4}{R} = \sin \alpha$



$R = \sqrt{3^2 + 4^2} = 5$

$\alpha \approx \tan^{-1}\left(\frac{4}{3}\right) \approx 53^\circ$ (nearest degree)

$\therefore 3\cos x + 4\sin x \approx 5 \cos(x - 53^\circ)$

(ii) $3\cos x + 4\sin x = 4$

$5 \cos(x - 53^\circ) = 4$

$\cos(x - 53^\circ) = \frac{4}{5}$

$-53^\circ \leq x - 53^\circ \leq 307^\circ$

$\therefore x - 53^\circ \approx -37^\circ, 37^\circ$ ✓

$x \approx 16^\circ, 90^\circ$ ✓

(d) $x = \tan^{-1}\left(\frac{7\pi}{8}\right)$

(i) $\tan\left(2 \times \frac{7\pi}{8}\right) = \frac{2 \tan\left(\frac{7\pi}{8}\right)}{1 - \tan^2\left(\frac{7\pi}{8}\right)}$ ✓

$-1 = \frac{2t}{1-t^2}$

$\therefore t^2 - 2t - 1 = 0$

(ii) $t = \frac{2 \pm \sqrt{4+4}}{2} = 1 \pm \sqrt{2}$ ✓

but $\tan\left(\frac{7\pi}{8}\right) < 0$ (2nd Quadrant)

$\therefore t = 1 - \sqrt{2}$ ✓

(10) (a) (i) $\ln\left(\frac{(x+1)^2}{2x}\right) = 2\ln(x+1) - \ln 2x$ ✓
 $= 2\ln(x+1) - \ln x - \ln 2$

(ii) $\frac{dy}{dx} = \frac{2}{x+1} - \frac{1}{x} - 0$ ✓

st. pt. at $\frac{dy}{dx} = 0 = \frac{2}{x+1} - \frac{1}{x}$

$$\frac{1}{x} = \frac{2}{x+1}$$

$$x+1 = 2x$$

$$\underline{x = 1}$$

$$y = \ln\left(\frac{2^2}{2}\right) = \ln 2$$
 ✓

∴ st. pt. at $(1, \ln 2)$

(b) RTP: $1 \times 4 + 2 \times 5 + 3 \times 6 + \dots + n(n+3) = \frac{1}{3}n(n+1)(n+5)$

proof: if $n=1$, LHS = 4

$$RHS = \frac{1}{3} \times 1 \times 2 \times 6 = 4 = LHS$$
 ✓

∴ The result is true for $n=1$.

Let's assume the result is true for some integer k .

i.e. $1 \times 4 + 2 \times 5 + 3 \times 6 + \dots + k(k+3) = \frac{1}{3}k(k+1)(k+5)$

$$\Rightarrow 1 \times 4 + 2 \times 5 + \dots + k(k+3) + (k+1)(k+4)$$
 ✓

$$= \frac{1}{3}k(k+1)(k+5) + (k+1)(k+4)$$

$$= \frac{1}{3}(k+1) \left[\cancel{\frac{1}{3}k(k+5)} + 3(k+4) \right]$$

$$= \frac{1}{3}(k+1) [k^2 + 8k + 12]$$

$$= \frac{1}{3}(k+1)(k+6)(k+2)$$
 ✓

$$= \frac{1}{3}(k+1)[(k+1)+1][(k+1)+5]$$

∴ If the result is true for k , then it's also true for $k+1$.

Since the result is true for 1, by the principle of Mathematical Induction, it's true for all positive integers

10
(C)

From the description, the line $y = 2 - x$ is the chord of contact on the parabola from the point P.

Let P be (x_0, y_0)

Eqn of chord of contact is

$$xx_0 = 2a(y + y_0)$$

$$x^2 = 6y \text{ has a focal length of } \frac{6}{4} = \frac{3}{2}$$

$\therefore$ Eqn is

$$xx_0 = 3(y + y_0)$$

$$\frac{xx_0}{3} - y_0 = y$$

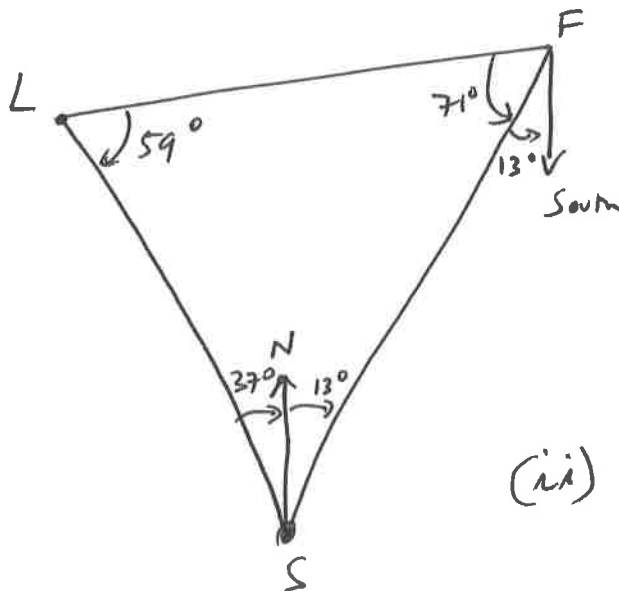
which is the same line as $y = 2 - x$

$$\therefore \frac{x_0}{3} = -1 \quad \text{and} \quad y_0 = -2$$

$$x_0 = -3$$

$\therefore$ The point P is $(-3, -2)$.

(d)



$$(i) \angle NSL = 360^\circ - 323^\circ = 37^\circ$$

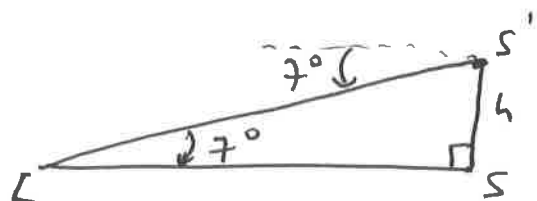
$$\angle LSF = 37^\circ + 13^\circ = 50^\circ$$

$$\angle SFL = 13^\circ \text{ (alt } \angle \text{S)}$$

$$\angle SFL = 264^\circ - 180^\circ - 13^\circ = 71^\circ$$

$$\therefore \angle FLS = 180^\circ - 50^\circ - 71^\circ \text{ (sum } \Delta) = 59^\circ$$

(ii)



$$\frac{h}{SL} = \tan 7^\circ \Rightarrow SL = \frac{h}{\tan 7^\circ}$$

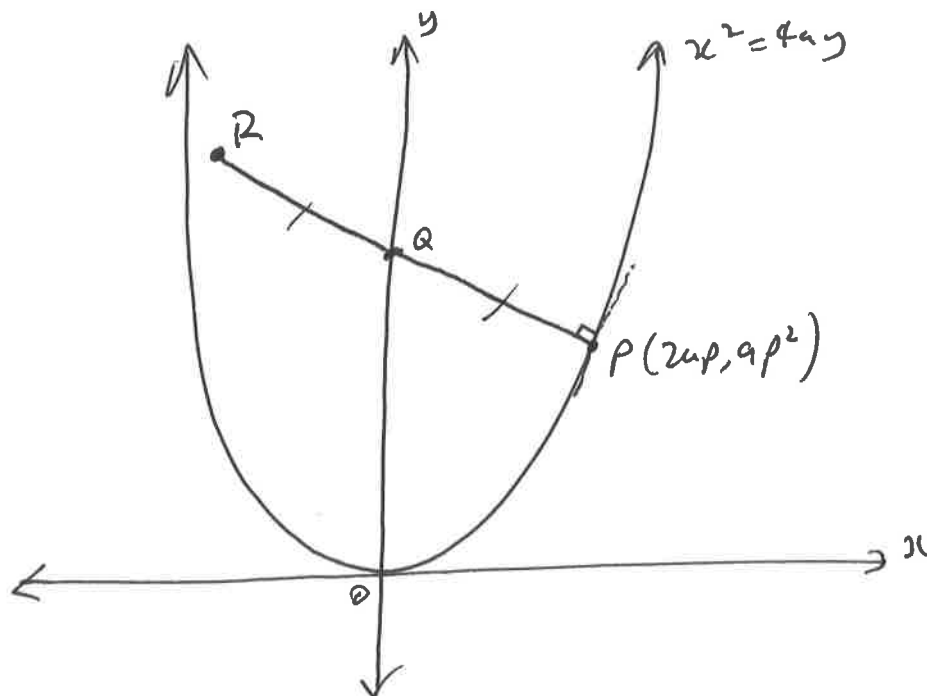
(iii) SINE RULE IN ΔFLS :

$$\frac{h}{\tan 7^\circ} = \frac{30.0 \text{ km}}{\sin 59^\circ}$$

$$h = \frac{30 \tan 7^\circ \sin 71^\circ}{\sin 59^\circ} \text{ km} \approx \underline{\underline{4.1 \text{ km}}}$$

11

(A)



(i)

$$x^2 = 4ay$$

$$\frac{dy}{dx} = \frac{2x}{4a} \quad \text{at } P, x = 2ap$$

$$\therefore m_{\text{tangent at } P} = \frac{2(2ap)}{4a} = p$$

$$\therefore m_{\text{normal at } P} = -\frac{1}{p}$$

$\therefore$ EQN normal:

$$y - (ap^2) = -\frac{1}{p}(x - 2ap)$$

$$py - ap^3 = -x + 2ap$$

$$x + py = ap^3 + 2ap \quad \checkmark$$

(ii) Q is the point on $x + py = ap^3 + 2ap$ where $x = 0$

$$\therefore py = ap^3 + 2ap$$

$$y = ap^2 + 2a$$

(assuming $p \neq 0$, else Q would not be defined)

$$\therefore Q \text{ is } (0, ap^2 + 2a) \quad \checkmark \text{ and } P \text{ is } (2ap, ap^2)$$

So since Q is the midpoint of R(x_0, y_0)

$$(0, ap^2 + 2a) = \left(\frac{x_0 + 2ap}{2}, \frac{y_0 + ap^2}{2} \right)$$

$$x_0 = -2ap, \quad y_0 \neq ap^2 = 2ap^2 + 4a$$

$$y_0 = ap^2 + 4a$$

$$\therefore R \text{ is } (-2ap, ap^2 + 4a) \quad \checkmark$$

(iii) Locus of R: $x^2 = 4a^2p^2 = 4a(ap^2)$ $\checkmark$
 $= 4a(y - 4a)$ $\checkmark$

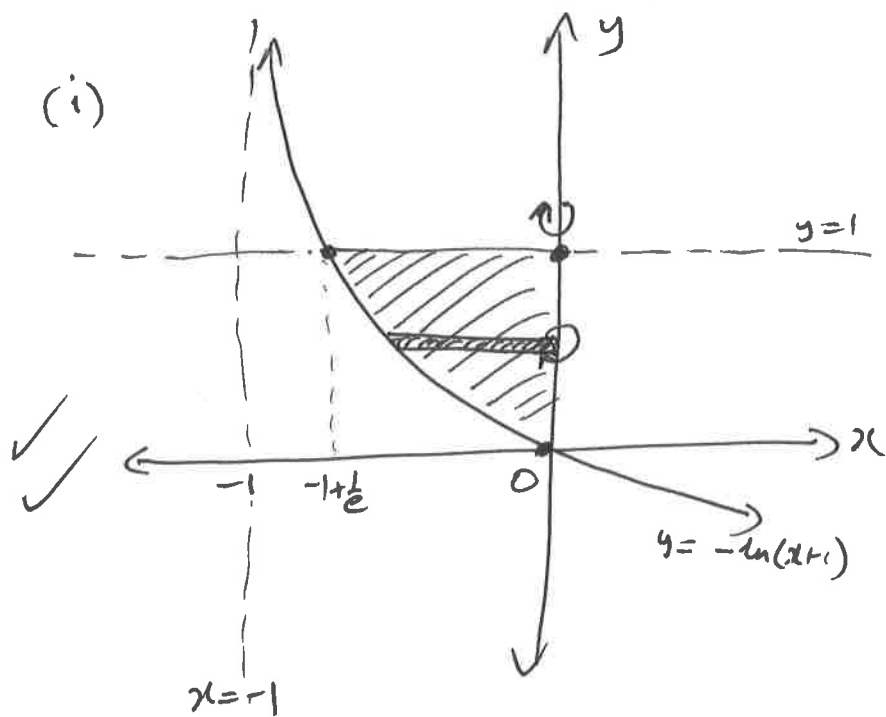
(iv) The locus of R is the translation of $x^2 = 4ay$ up by $4a$ units, but not including the vertex $(0, 4a)$ since $p \neq 0$. $\checkmark$

① (b) $y = -\ln(x+1)$ (i)

$$e^{-y} = x+1$$

$$x = e^{-y} - 1$$

when $y=1$, $x = \frac{1}{e} - 1$



(ii) $V = \pi \int_{y=0}^{y=1} (x^2) dy$ ✓

$$= \pi \int_0^1 (e^{-y} - 1)^2 dy$$

$$= \pi \int_0^1 (e^{-2y} - 2e^{-y} + 1) dy$$
 ✓

$$= \pi \left[\frac{e^{-2y}}{-2} - \frac{2e^{-y}}{-1} + y \right]_0^1$$

$$= \pi \left[\left(-\frac{e^{-2}}{2} + \frac{2}{e} + 1 \right) - \left(-\frac{1}{2} + 2 + 0 \right) \right]$$

$$= \pi \left(-\frac{1}{2e^2} + \frac{2}{e} - \frac{1}{2} \right)$$

$$= \frac{\pi}{2e^2} (-e^2 + 4e - 1) \quad (\approx 0.528)$$

units³